Md Modassir Firdaus

DOB : 02/01/1993

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Education

PhD Candidate in Mechanical Engineering Jan-2020 to Ongoing

Indian Institute of Technology Gandhinagar (IITGN),
Visiting PhD Scholar, BART Lab Mahidol University

Thesis: "*Virtual Actuation Framework for Kinematics and Control of Tendon-Driven Continuum Robots with Cadaveric Validation in Colon Navigation*" Thesis Submitted

Thesis Adviser: Prof. Madhu Vadali (Mechanical Engineering IITGN, India)

Surgical Robot Development Adviser: Prof. Jackrit Suthakorn (BART Lab Mahidol University, Thailand) and Prof. Branesh M. Pillai (AIT, Thailand)

Clinical Trial Adviser: Prof. Chumpon Wilasrusmee (Minimally Invasive Surgeon, Ramathibodi Hospital, Thailand)

Bachelor's in mechanical engineering July 2012- April 2016

Jawaharlal Nehru Technological University
75.4 %, Best CGPA of the batch

Research Experience

Research Associate I Oct 2025 – Present

Indian Institute of Technology Gandhinagar, India

PhD Researcher (Junior Research Fellow → Senior Research Fellow) Jan 2020 – Oct 2025

Indian Institute of Technology Gandhinagar, India

Teaching Experience

Founder and Faculty of Physics and Mathematics May 2016 -Dec 2019

Firdaus Academy for JEE Main & Adv.

- I have learned to manage an institute.
- Leading a team of faculties to fulfill the objective.

Learn to deal with and manage the challenges created by students to a different extent.

Publications

[J1]	**Md Modassir Firdaus**, Omkar Dilip Paranjape, and Madhu Vadali	Finite Element Method-Based Dynamic Modeling Framework for Flexible Continuum Manipulators	ASME Journal of Mechanisms Robotics [Q1]
[C2]	**Md Modassir Firdaus** and Madhu Vadali	Virtual Tendon-Based Inverse Kinematics of Tendon-Driven Flexible Continuum Manipulators	AIR '23: Proceedings of the 2023 6th International Conference on Advances in Robotics, ACM [IF 1.3]
[C2]	**Md Modassir Firdaus**, Smit Parjapati, and Madhu Vadali	Initial State Update Based Position Control of Flexible Continuum Manipulators Utilizing Virtual Actuation Space	IEEE International Conference on Robotics and Biomimetics (IEEE-ROBIO 2024) [IF 1.81]

[J2]	**Md Modassir Firdaus**, Shail Jadav, and Madhu Vadali	Trajectory Tracking of Multi-Section Tendon-Driven Flexible Continuum Manipulators using Virtual Actuation Space Control.	Robotica [IF 2.7]
[C3]	**Md Modassir Firdaus**, Vikranth Mallru, Harsh Malodia, and Madhu Vadali	Bioinspired Rolling-Disk Continuum Robot: Logarithmic Spiral and Constant Curvature Design with Contraction Capabilities	**ICRA 2026** Accepted
[J3]	**Md Modassir Firdaus**, B. M. Pillai, Madhu Vadali, C. Wilasrusmee and J. Suthakorn	Design and Cadaveric Evaluation of a Low-Cost, 3D-Printed Tendon-Driven Continuum Endoscope for Colon Intervention: A Preliminary Study	IEEE Medical Robotic and Bionics (Under Review)
[J4]	**Md Modassir Firdaus** and Madhu Vadali	Topology Informed Point Claud Based Path Planning Using RRT* for Anatomical Cavity and Different Topological Space.	IJRR (Under Review)

Patent

1. Vadali, Madhu and **Firdaus, Md Modassir**, "A system for performing minimally invasive surgery (MIS)", Indian Patent Office, Patent Application No.: 202421002550, (Published)
2. **Md Modassir Firdaus**, Harsh Malodia, and Madhu Vadali, "A Rolling Disk Embodied Design of Continuum Robot" || Patent Application No: 202621020127 Date: 20.02.2026
3. **Md Modassir Firdaus,** and Madhu Vadali, "Ergonomic Flexible Minimally Invasive (ErgoFlex MIS) Instrument with Wrist-Aligned Gimbal Actuation" || P-600779 (Filing)

Awards

[A1]	2025	**Winner of MIT Media Lab AI and Robotics Hackathon 2025**
[A2]	2025	**Gandhian Young Technological Innovation (GYTI) Award 2024, for the development of "Flexible Robot for Transanal Minimally Invasive Intervention"** (One of the most prestigious awards for young innovators)
[A3]	2024	Overseas Research Exposure Fellowship by Indian Institute of Technology Gandhinagar
[A4]	2024	Mahidol University Residence Fellowship
[A5]	2024	Grant for Attending International Conference
[A6]	2024	Winner Regional Finale of Boeing BUILD
[A7]	2014	Winner college level line following mobile robot competition.

Skills

Control Dynamixel and Rigid Linked Robots like **xArm6**, **Adverb Heal**
Designing the continuum robot using **CAD**

Sensors, Actuators and Controller Integration (Electronic Setting)
Worked with **NI Based Real-Time Controller** and **DAQ**

Latex, Graphic Design

Motion Capture, Continuum Robot and Adverb Heal Integration using **ROS 1**
MATLAB for Mathematical Calculation and Simulations
Python and **Lab View** for Hardware Implementations
CNC, Lathe, Leaser Cutting, 3D Printing and other Manufacturing Tools for building experimental setup and prototypes

Talk and Seminar Delivered

IEEE Medical Robotics Seminar, Mahidol University, Thailand — Nov 06, 2024

Invited Speaker AI- Robotics Conference Dialogue on Healthcare MBZUAI Abu Dhabi — April 23, 2025- April 25, 2025

Experience as Reviewer

- IEEE APSCON
- IEEE RAL
- International Conference on Advances in Robotics
- i-Create 2025

Teaching Assistant Experience

Graduate Teaching Fellow [Manufacturing] Indian Institute of Technology Gandhinagar	I taught undergraduate course on **Manufacturing Technology I**, Prof. Madhu Vadali
Teaching Assistant Indian Institute of Technology Gandhinagar	**Control Theory**, Prof. Madhu Vadali
Teaching Assistant Indian Institute of Technology Gandhinagar	**ME LAB II** learning-by-doing, Prof. Harish PM
Teaching Assistant Indian Institute of Technology Gandhinagar	**Manufacturing Technology II**, Prof. Madhu Vadali
Teaching Assistant Indian Institute of Technology Gandhinagar	**Control Theory**, Prof. S. Rajendaran

Relevant Course Work

Introduction to Robotics	Nonlinear Control
Control Theory	Mechanics of Rod
Modern Control Theory	Digital Control Systems

Hobbies

Playing Squash| Swimming, Represented IIT Gandhinagar in inter-IIT swimming championship, Free-Style |

Some Extra Activity

Karate Yellow Belt | Annual Training Camp NCC, Army Wings